

10 Clinical Applications

A Comprehensive Guide to the Practical and Clinical Uses of 3D Printing in Dentistry



Chapter 1: The Versatile Dental Printer

One of the greatest advantages of 3D printing is that a single hardware unit can produce a wide array of clinical devices. By simply switching the resin vat and material type, a dentist can transition from printing diagnostic models to printing surgical guides, biocompatible splints, or temporary crowns.

This flexibility maximizes the return on investment (ROI) for the dental clinic, allows for immediate in-office modifications, and cuts out the shipping delays associated with traditional external dental laboratories.

Below, we outline the first 4 of the top 10 most common clinical applications of 3D printing in modern practices:

1. Diagnostic Study Models

Printed using standard model resin. Allows clinicians to analyze patient bite alignment, explain treatment paths, and keep permanent physical records of cases.

2. Clear Aligner Thermoforming Models

Models printed from orthodontic planning software showing gradual tooth movement. Clear aligners (such as Align-All) are then thermoformed directly over these printed models.

Chapter 2: Surgical & Splint Applications

The use of 3D printing has dramatically improved safety and predictability in implantology and restorative dentistry. Here we look at applications 3 through 6:

3. Surgical Drill Guides

Printed using highly rigid, sterilizable, biocompatible resins. These guides slide over the patient's teeth during surgery and have metal sleeves that guide the implant drill at the precise angle and depth mapped in the CT scan.

4. Indirect Bonding Trays (IBT)

Printed using flexible biocompatible resins. Orthodontists place brackets into the slots of the printed tray, align the tray on the patient's teeth, and light-cure all brackets simultaneously, cutting appointment times in half.

5. Occlusal Guards & Splints

Printed from clear, high-impact biocompatible materials. Used to protect teeth in patients who suffer from bruxism (teeth grinding) and to alleviate TMJ joint disorders.

6. Anatomical Maxillofacial Models

Printed from CT scans to show complex bone fractures or impacted teeth. Allows surgeons to rehearse procedures and pre-bend plates before the patient enters the operating room.

Chapter 3: Restorative & Prosthetic Applications

3D printing is now actively replacing traditional milling and casting methods for producing final oral prosthetics. We look at applications 7 through 10:

7. Temporary Crowns & Bridges

Printed with highly aesthetic, tooth-colored restorative resins. Placed directly in the mouth for short-term use while a permanent ceramic crown is milled.

8. Custom Impression Trays

Custom-shaped trays designed specifically for the patient's mouth. These are printed to ensure optimal tray fit and material thickness during final impression capture.

9. Digital Full Dentures

Printed in two parts: a pink gingiva-colored base and tooth-colored denture arches, which are then bonded together. This speeds up denture fitting and makes replacement simple.

10. Castable Pressing Patterns

Printed using zero-ash burnout resins. Used to create detailed patterns for pressing ceramic veneers or casting metal frameworks, bypassing manual waxing.

Chapter 4: Clinical Summary & Benefits

By incorporating these applications into daily workflows, dental practices unlock several transformative benefits:

- Reduced Treatment Cost: Raw resin printing costs a fraction of physical shipping and plaster cast storage.
- Chairside Convenience: Make modifications and reprint models while the patient is still in treatment, reducing follow-up visits.
- Customization: Custom orthodontic trays, splints, and guides fit the patient's anatomy perfectly, leading to higher comfort.
- Seamless Records: All patient models are stored digitally as STL files, eliminating physical model cabinets.

Future Outlook: As material science advances, direct printing of permanent ceramic crowns and shape-memory clear aligners will soon become standard, further reducing laboratory steps.